OPERATIONS RESEARCH

The Science of Better Decisions

How mathematical modeling, optimization, and structured decision-making turn complex, resource-constrained problems into solvable ones — across manufacturing, logistics, healthcare, and beyond.

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FOUNDATIONS

What Is Operations Research?

Optimization is the art of finding the best alternative(s) among a given set of options — the process of finding the maximum or minimum value a given function can attain within its domain of definition.

Operations Research (OR) applies this idea at scale: it is the discipline of using mathematical modeling, statistical analysis, and algorithmic methods to arrive at optimal or near-optimal decisions in complex, resource-constrained systems — turning a vague goal like “do this efficiently” into a precisely defined, solvable problem.

Analyze

Understand the system and its constraints

Model

Represent it mathematically

Optimize

Search the solution space

Decide

Act on a defensible best choice

FOUNDATIONS

Why Operations Research Matters

The same mathematical machinery — objective functions, constraints, and search — recurs everywhere a limited resource must be allocated well.

Manufacturing

Scheduling, layout, and product mix

Logistics

Routing, distribution, facility location

Healthcare

Staff rostering, bed allocation, triage

Finance

Portfolio and risk optimization

Aerospace

Fleet scheduling and crew assignment

A single percentage-point improvement in routing efficiency, inventory turnover, or scheduling utilization compounds into enormous savings at industrial scale — which is precisely why OR sits at the center of modern operations management.

PART ONE

The Taxonomy of Optimization Problems

Every OR problem is first classified along two axes: whether its variables and outcomes are fixed or uncertain, and whether its decision space is continuous or discrete.

TAXONOMY

Deterministic vs. Stochastic — the First Split

Deterministic Optimization

All data — costs, capacities, demand — is known with certainty. The challenge is purely computational: search a well-defined space for the optimum.

- Continuous Optimization (convex, concave)
- Combinatorial Optimization (LP, IP, assignment, knapsack)
- Routing & Scheduling (TSP, VRP, job scheduling)

Stochastic Models

Randomness is intrinsic — arrival times, demand, or travel times follow probability distributions. Solutions must perform well on average or with a stated confidence level.

- Network & Flow Models (transportation, network flow)
- Inventory Control Problems
- Queueing Theory

TAXONOMY · DETERMINISTIC

Continuous Optimization

When decision variables can take any value within a range, the shape of the objective function determines how tractable the problem is.

Convex Optimization

A single global optimum; local search methods (gradient descent, interior-point) are guaranteed to find it. The workhorse of modern large-scale optimization.

Concave Optimization

Maximizing a concave function is the mirror image of convex minimization — common in utility maximization and resource-allocation problems.

Combinatorial Optimization

Here the decision variables are discrete — choose a subset, an ordering, or an assignment from a finite (but often astronomically large) set of options.

Linear Programming (LP)

Linear objective, linear constraints

Integer Programming (IP)

LP with integer-valued variables

Assignment Problem

Match agents to tasks at minimum cost

Knapsack Problem

Maximize value under a capacity limit

Travelling Salesman (TSP)

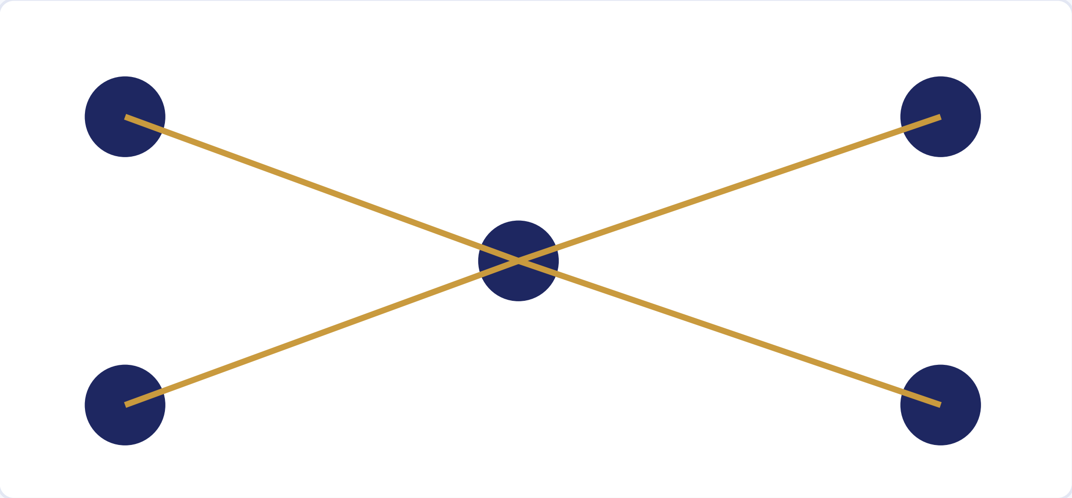
Shortest tour visiting every city once

Vehicle Routing (VRP)

Multi-vehicle delivery route optimization

Network & Flow Models

Many logistics and resource-distribution problems reduce to moving a quantity through a network of nodes and arcs at minimum cost or maximum throughput.



Source → hub → sink flow

Transportation Problem

Minimum-cost shipment from multiple supply points to multiple demand points, respecting capacity and requirement constraints.

Network Flow Problems

Inventory Control & Queueing Theory

Inventory Control Problems

When and how much to reorder under uncertain demand and lead time, balancing holding cost against stockout risk. Classic models: EOQ, (s, S) policies, newsvendor problem.

Queueing Theory

Models of waiting lines — arrivals, service times, and server capacity — used to size staffing, checkout counters, call centers, and computer-network buffers.

PART TWO

The Optimization Workflow

Solving a real OR problem is rarely a single step. It is an iterative process spanning problem formulation, handling real-world imperfection, algorithm choice, refinement, and implementation.

STAGE

1

Formulate before you compute.

Problem Formulation & Modeling

Every optimization begins by translating a messy real-world goal into a precise mathematical object.

Objective Identification

What quantity are we actually maximizing or minimizing?

Decision Variables

What can we actually choose or control?

Constraint Formulation

What limits — capacity, budget, time — must be respected?

Mathematical Representation Choice

Linear? Nonlinear? Integer? The choice shapes every step after.

STAGE

2

Models meet messy reality.

Real-World Imperfections & Uncertainties

No dataset is perfect and no future is fully known — a model has to survive contact with both.

Data Quality Handling

Missing values, measurement error, and inconsistent sources

Uncertainty Incorporation

Stochastic parameters, scenario analysis, robust optimization

STAGE

3

Match the method to the problem.

Algorithm Selection & Customization

No single algorithm dominates every problem class — selection is itself an engineering decision.

Problem Characteristics Understanding

Convexity, scale, and structure dictate what will even converge

Algorithm Suitability

Exact methods vs. metaheuristics vs. specialized solvers

Parameter Tuning

Step sizes, population sizes, cooling schedules — tuned, not guessed

STAGE

4

Solutions are refined, not revealed.

The Iterative Process

Optimization is rarely a single pass — it is a disciplined loop of solve, check, and adjust.

- **Continuous Refinement**

  Re-solving as new information or constraints emerge

- **“Good Enough” Recognition**

  Knowing when further search no longer justifies its cost

STAGE

5

A number is not a decision.

Interpreting & Implementing Solutions

The optimizer's output only becomes valuable once it is understood, trusted, and acted upon by people.

- **Solution Sensitivity**

  How much does the answer change if inputs shift slightly?

- **Effective Communication**

  Translating results for non-technical decision-makers

- **Qualitative Factors & Human Judgment**

  Optimal on paper is not always feasible in practice

PART THREE

Operations Research in Production Management

A single application domain — production management — already touches manufacturing, maintenance, project scheduling, and physical distribution.

Manufacturing & Workforce Planning

Production Scheduling & Sequencing

Ordering jobs across machines to meet due dates

Stabilization of Production

Smoothing output despite demand fluctuation

Employment — Training & Layoffs

Workforce sizing decisions tied to production plans

Optimum Product Mix

Which products, in what quantities, maximize contribution

Maintenance & Project Scheduling

Maintenance

- Maintenance Policies
- Preventive Maintenance

Project Scheduling

- Allocation of Resources
- Critical-path & timeline planning

Physical Distribution & Facilities Planning

Location & Size

Warehouses, distribution centres, retail outlets

Distribution Policy

How stock moves from source to point of sale

Facility Numbers & Location

Factories, warehouses, hospitals — how many, and where

Loading, Unloading & Transport

Railways, trucks, and transport schedules

WHY THIS MATTERS TO MY RESEARCH

Operations Research gives me the formal language to turn engineering intuition — about defect tolerance, layout efficiency, or process parameters — into models that can be optimized, validated, and defended quantitatively.

Thank you

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